Predicting NBA Player Injury Risk from In-Season Workload Patterns

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Project 1, Milestone 3



Business Problem

NBA teams invest heavily in load management and sports science staff, yet they lack a simple, data-driven way to flag which specific players are entering a higher-risk window based on their own recent workload.

The NBA's own analytics group examined roughly 150 players per season from 2014 to 2023 and found no significant correlation between game-participation frequency, schedule density, or cumulative participation and injury.

Research Questions

1. Can recent workload predict elevated risk in the games that follow?
2. Which workload features are the strongest predictors: minutes, rest, or schedule density?
3. Do the findings support or contradict the NBA's own conclusion on schedule density?

The goal is to create a model that complements existing load-management practices for team performance staff, rather than replacing their judgement.



Background

Injury risk in professional basketball has become an increasingly prominent analytics topic as teams weigh competitive performance against long-term player health. There are several prior works that inform this project:

- Bontemps, T. (2024, January 13). *NBA report: No link between load management, less injury risk*. ESPN. https://www.espn.com/nba/story/_/id/39288379/nba-report-no-link-load-management-less-injury-risk
- Bullock, G. S., Ferguson, T., Vaughan, J., Gillespie, D., Collins, G., & Kluzek, S. (2021). Temporal trends and severity in injury and illness incidence in the National Basketball Association over 11 seasons. *Orthopaedic Journal of Sports Medicine, 9*(6). https://doi.org/10.1177/23259671211004094
- Torres-Ronda, L., Gámez, I., Robertson, S., & Fernández, J. (2022). Epidemiology and injury trends in the National Basketball Association: Pre- and per-COVID-19 (2017–2021). *PLOS ONE, 17*(2), e0263354. https://doi.org/10.1371/journal.pone.0263354
- Wang, E. (2025). *Using injury-risk forecasting to quantify financial impact in the NBA*. Wharton School, University of Pennsylvania. https://wsb.wharton.upenn.edu/wp-content/uploads/2025/12/WANG_Using-Injury-Risk-Forecasting-to-Quantify-Financial-Impact-in-the-NBA-1.pdf

Three data sources were combined for this analysis, all sourced from Kaggle:

- NBA Injuries (2010 – 2020): 27,105 raw transaction records scraped from Pro Sports Transactions, with fields for date, team, player acquired, player relinquished, and a free-text notes field describing the transaction: https://www.kaggle.com/datasets/ghopkins/nba-injuries-2010-2018
- NBA Games: 26,651 game-level records (2003 – 2022) with final scores, shooting statistics, and a season label for each game: https://www.kaggle.com/datasets/nathanlauga/nba-games
- NBA Games Details: 668,628 player-level box score records (one row per player per game), including minutes played, points, and other box-score statistics, joined to game dates and seasons: https://www.kaggle.com/datasets/nathanlauga/nba-games



Data Explanation

Turning free-text transactions into a reliable label

Category	Rows	Action
Raw placed_on_IL entries	17,560	Starting point
Non-injury absences (rest, personal, conditioning)	620	Reclassified
Coach/staff entries	14	Reclassified
Malformed team-as-player entry	1	Reclassified
Final injury events used for modeling	16,935	Retained

Name normalization raised the player-match rate between datasets from 94.3% to 97.0%.



Methods

Five workload signals:

- Rolling average minutes (trailing 5 and 10 games)
- Cumulative minutes played this season
- Days of rest since last game
- Team-level back-to-back flag
- Back-to-backs in the trailing 5 games

Target: injured_next_7d

A forward-looking binary label: 1 if the player's next confirmed injury occurs within 7 days after this game. Every feature uses only information known before the game, nothing leaks from the outcome being predicted.

Time-based split, two models, imbalance-aware

Train / Test Split:
Train: 2010 – 2017 seasons
Test: 2018 – 2019 season

Split by season so the model is never evaluated on games that happened before its training data

Two Classifiers:

Logistic Regression (interpretable baseline) and Random Forest (primary model), both using class_weight = balanced

Evaluation:

Precision, recall, F1, and ROC-AUC. Not raw accuracy, given roughly 1 injury-preceding game per 11 healthy ones.



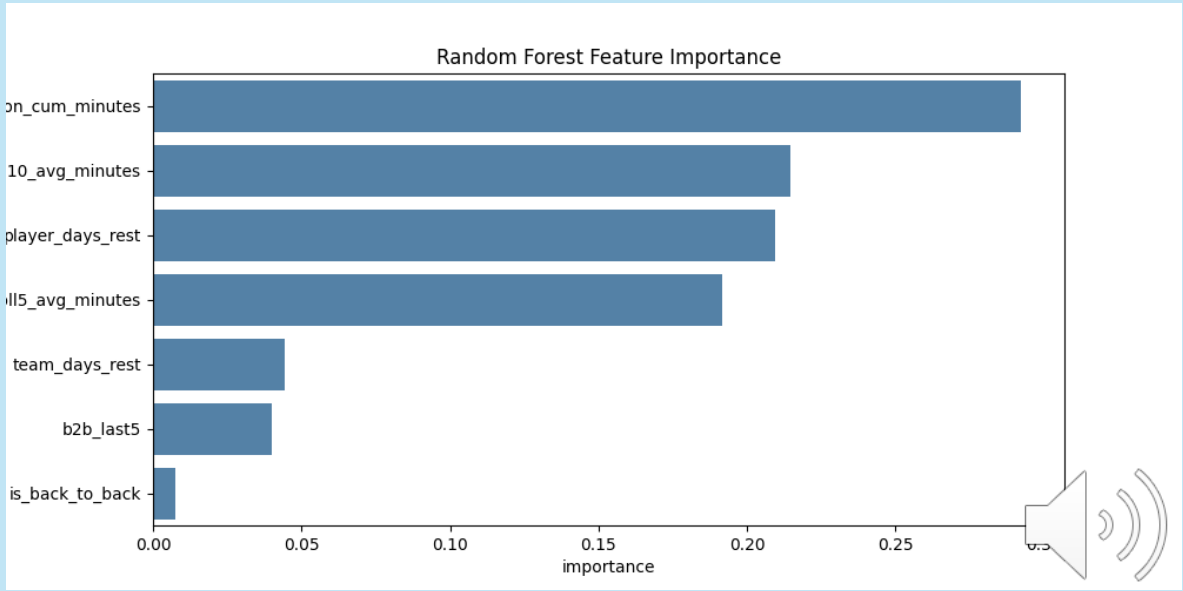
Analysis

Model performance on the held-out test set

Metric	Logistic Regression	Random Forest
Precision (Injury class)	0.11	0.11
Recall (Injury class)	0.48	0.59
F1-score (Injury class)	0.18	0.19
Overall Accuracy	0.61	0.55
ROC-AUC	0.569	0.587

Test-set positive rate: 9.0% of games precede an injury withing 7 days
Training-set positive rate: 8.1%

What's driving the model's predictions?



Conclusion

A weak, but legitimate signal

RQ1: Recent workload carries detectable but weak signal (AUC 0.57 – 0.59), well below the 0.70 threshold generally considered clinically useful.

RQ2: Cumulative workload and rest patterns matter most; back-to-back scheduling contributes comparatively little.

RQ3: Findings reinforce the NBA's own conclusion that schedule density has little direct link to injury.

